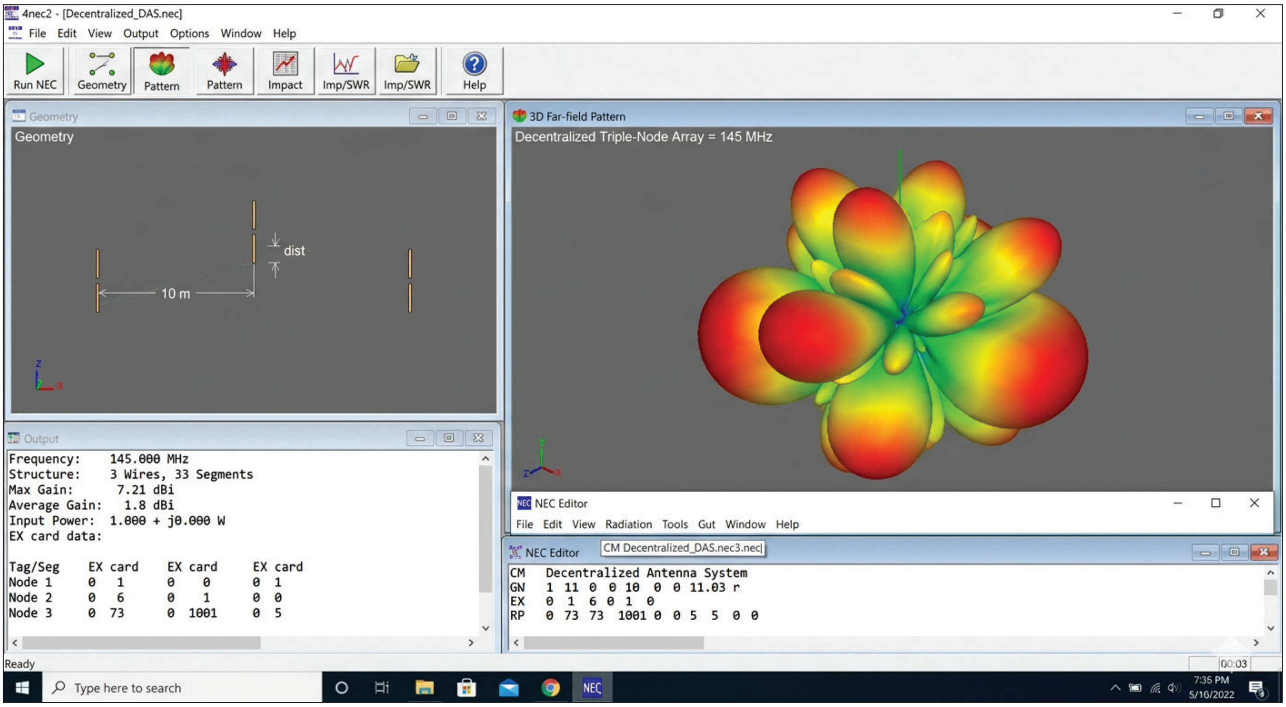


Figure 3: Decentralised antenna simulation in 4nec2 simulator

**Table 2: Key features of 4nec2 simulator**

| Smith chart | Time and frequency analysis | Time plot |
|---|---|---|
| Impedance matching | Linear far-field plot | Gain, F/B and F/R ratio |
| 3D near-field pattern | 3D viewer | Area coverage |
| Compare patterns | Current distribution | Current distribution |
| Distance plot | FF surface plot | Far-field pattern |
| Field analytics | Optimiser window | Geometry edit |
| Geometry builder | NEC editor | Helix model |
| SWR for freq-sweep | Search space | Ship model |

decentralised solutions the dedicated small node is in each home with the option to share the resources and bandwidth on demand or dynamically.

GNU Radio (Figure 2) can be used for simulating radio signal analytics when the blockchain is being integrated. It is available as a free and open source platform on *https://www.gnuradio.org.* Its implementations include 2G, 3G, 4G, LTE, RFID, and 5G.

Decentralised antenna design is a key research area in Cell Free Massive MIMO (CF-mMIMO). In general implementations, there is a specific antenna connected to the devices. In blockchain-based decentralised antennas, multiple nodes at multiple locations coordinate with each other to provide the signals.

A decentralised antenna can be simulated using the 4nec2 simulator that is available under free and open source distribution at *https://www.qsl.net/4nec2/.* The results of the simulation on 4nec2 help predict the outcomes of blockchain-based decentralised antenna design.

Free and open source simulators can be used by researchers for decentralised scenarios so that blockchain-based implementations can be analysed before deployment on site. END

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